

B.2.2 Exam Practice 2: Networking

Date: 4/27/2026, 3:03:57 AM

Time Spent: 11:43

Score: 83%

Passing Score: 80%



Question 1

✔ Correct

A remote support technician is speaking with a new rural business client who has issues with their internet connection. As the remote technician gathers information for a local technician to visit the site, the user mentions that the local tech referred to their connection as the "local loop" and the "last mile." What device would the remote technician conclude that they use to connect to the internet?

- ☐ Optical network terminal
- ☒ DSL modem ✓ Correct
- ☐ Wi-Fi access points
- ☐ Cable modem

Explanation

A digital subscriber line (DSL) uses higher frequencies in copper telephone lines from the "last mile" or "local loop" of plain old telephone systems (POTS) as a communications channel. A DSL modem interfaces between telephone lines and local networking devices, providing routing and internet connection.

A cable modem is a cable-internet-access digital modem using a coaxial connection to the service provider's fiber-optic core network. This connection is not typical in rural areas.

Wi-Fi access points provide connectivity for devices to the internet; they depend on local routing for internet connection. Modems typically do this routing.

An optical network terminal (ONT) converts between optical and electrical signaling to facilitate full fiber internet connection types. This is not typical in rural communities.

Related Content

 6.1.1 Internet Connection Types and Modems

 6.1.2 Digital Subscriber Line Modems

resources\questions\pt_core1_220_1201_2_5_002.question.xml

Question 2

— Partial

Which well-known mailbox access protocols utilize mailbox servers to allow the user's client email software to retrieve messages from the mailbox? (Select two.)

☒ POP3 ✓ Correct☒ FTP ✗ Incorrect☐ DNS☐ IMAP ✓ Correct**Explanation**

The Post Office Protocol (POP) is an early example of a mailbox access protocol. POP is often referred to as POP3 because the active version of the protocol is version 3.

The Internet Message Access Protocol (IMAP) is a mail retrieval protocol. Still, its mailbox management features support permanent connections to a server and connect multiple clients to the same mailbox simultaneously.

The File Transfer Protocol (FTP) allows clients to upload and download files from a network server. It uploads files to websites.

Domain Name System (DNS) allows clients to request resources using plain names rather than IP addresses.

Related Content 7.1.5 Mail Servers 7.1.6 Mailbox Servers

resources\questions\pt_core1_220_1201_2_3_003.question.xml

Question 3

✔ Correct

What characteristics of Wi-Fi 5 (802.11ac) allow for increased bandwidth and support for modern networks? (Select three.)

- ☐ Designed to work in both the 2.4 GHz and 5 GHz bands
- ☒ Designed to work only in the 5 GHz band ✔ Correct
- ☒ Typically supports up to four simultaneous clients ✔ Correct
- ☒ Downlink MU-MIMO ✔ Correct

Explanation

Wi-Fi 5 works only in the 5 GHz band. A dual-band access point can use its 2.4 GHz radio to support clients on legacy standards (802.11g/n), but it will not support Wi-Fi 5 on that 2.4 GHz band.

In Wi-Fi 5, downlink multi-user, multiple input, multiple output (DL MU-MIMO) allows the access point to use its multiple antennas to simultaneously send data to up to four clients.

Wi-Fi 5 allows up to eight streams, though most Wi-Fi 5 access points only support 4x4 streams in practice.

Wi-Fi 6 works in both the 2.4 GHz and 5 GHz bands. This is not a characteristic of Wi-Fi 5.

Related Content

 5.2.10 Power over Ethernet

 5.4.7 Wi-Fi 5 and Wi-Fi 6

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Question 4

✔ Correct

A user calls to report that they cannot connect to the internet and various other internal resources from their laptop. After a technician asks them what IP address their Windows laptop has, they report it has an address of 169.254.0.115. What problem is the user's laptop having?

- ☐ The laptop has a Class A private address.
- ☐ The laptop has a misconfigured static IP address.
- ☐ The laptop has a misconfigured default gateway address.

☒ The laptop cannot get an address from the DHCP server. ✔ Correct

Explanation

Automatic private IP addressing (APIPA) is a mechanism for Windows hosts to obtain an address automatically that cannot contact a DHCP server to revert to using an address from the range 169.254.x.y. This is also called a link-local address.

Using static addressing requires an administrator to visit each computer to enter the configuration information for that host manually. This IP would likely not be statically assigned.

10.0.0.0 to 10.255.255.255 is a Class A private address range.

The default gateway parameter is the IPv4 address of a router. If the laptop cannot contact a DHCP server, it will not get a default gateway at all.

Related Content

6.2.12 Static Versus Dynamic Host Address Configuration

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Question 5

✔ Correct

This protocol facilitates requests for an IP address and other network settings and uses ports 67 and 68.

- ☐ Network Basic Input/Output System
- ☐ Domain Name System
- ☐ Lightweight Directory Access Protocol
- ☒ Dynamic Host Configuration Protocol ✔ Correct

Explanation

Dynamic Host Configuration Protocol (DHCP) utilizes two ports. DHCP clients use UDP port 68 to request an IP address configuration from a server. The server responds using UDP port 67 to provide clients with an IP address configuration.

Domain Name System (DNS) facilitates the identification of hosts by name alongside IP addressing and is utilized over UDP traditionally or TCP port 53 in specific circumstances.

Network Basic Input/Output System (NetBIOS) supports legacy Windows versions' networking features that include name service for name registration and resolution, similar to DNS today. It utilizes UDP and TCP ports 137-139.

Lightweight Directory Access Protocol (LDAP) queries information about network users and resources. Third-party applications can use this to query Active Directory. LDAP uses TCP port 389.

Related Content[6.3.6 Well-Known Ports](#)[resources\questions\pt_core1_220_1201_2_1_005.question.xml](#)

Question 6

✔ Correct

This part of an IP address configuration allows a host to communicate with a remote network, most notably the Internet. Otherwise, the host will only communicate with the local network without this.

- ☐ Static address
- ☐ APIPA address
- ☒ Gateway address ✔ Correct
- ☐ DNS address

Explanation

The default gateway is the IPv4 router address, such as 192.168.0.1. This setting is not compulsory, but failure to enter a gateway would limit the host to only communicate on the local network.

DNS addresses provide resolution of host and domain names to their IP addresses and are essential for locating resources on the internet.

Using static addressing requires an administrator to visit each computer to enter the configuration information for that host manually.

When a host uses an APIPA address, it can communicate with other hosts on the same network using APIPA. Still, it cannot reach other networks or communicate with hosts who have obtained a valid DHCP lease.

Related Content[6.2.9 IPv4 Forwarding](#)[6.2.11 IPv4 Host Address Configuration](#)[resources\questions\pt_core1_220_1201_2_6_004.question.xml](#)

Question 7

✕ Incorrect

A new IT support technician troubleshoots a user's report of intermittent network connectivity. After inspecting the workstation, they see a label above the station's RJ-45 port wall connection. This label is likely to identify what device?

- ☐ Switch
- ☒ Hub ✕ Incorrect
- ☐ Patch panel ✓ Correct
- ☐ Firewall

Explanation

In most types of office cabling, the computer is connected to a wall port and, via cabling running through the walls, to a patch panel.

A switch is an intermediate system used to establish contention-free network segments at OSI layer 2 (Data Link). A port on a patch panel will connect to a switch to provide office network connectivity.

A hub is a legacy network hardware device that implemented the 10BASE-T and 100BASE-T Ethernet cabling designs. Hubs were used to provide multiple connections to computers in a small area and are inefficient at network management.

A basic firewall has rules that list source or destination network addresses and protocol types and whether to allow or block traffic that matches the rule.

Related Content

-  5.2.3 Patch Panels
-  5.2.6 Switches
-  5.2.9 Unmanaged and Managed Switches
-  5.5.1 Network Switches
-  15.2.4 Group Policy and Login Scripts

resources\questions\pt_core1_220_1201_2_5_004.question.xml

Question 8

✔ Correct

Users reported an incident where they could not connect to network shares on a specific file server but could utilize all other resources and the internet. After logging into the server locally to check, it seems to be up and connected to the network. There is no response after going to a user's station and using the ping command to contact the server by the documented IP address. What should be the issue to investigate next?

- ☒ Check the server's IP address. It may have a DHCP-provisioned address instead of a static address. ✔ Correct
- ☐ Ensure the server has a Class A IPv4 address.
- ☐ Check the user's default gateway to ensure it can communicate to remote networks.
- ☐ Check to see if the user workstation has been given an APIPA address.

Explanation

If the server locally appears online and connected, it has an address, but likely one provisioned from DHCP if workstations cannot ping the server by the documented IP address.

The user workstation would not likely reach all other resources if it had an APIPA address.

If the user station can connect to other resources, especially the internet, there is likely no issue with the gateway.

It may not be a Class A address, depending on the type of network structure. A technician can also use class B and C addresses for private addressing.

Related Content

resources\questions\pt_core1_220_1201_2_6_005.question.xml

Question 9

✔ Correct

Which application protocol underpins file and printer sharing on Windows networks and usually runs directly over the TCP/445 port?

- ☐ Network Basic Input/Output System
- ☐ File Transfer Protocol
- ☐ HyperText Transfer Protocol
- ☒ Server Message Block ✓ Correct

Explanation

Server Message Block (SMB) is the application protocol underpinning file and printer sharing on Windows networks. SMB usually runs directly over TCP port 445.

Network Basic Input/Output System (NetBIOS) allows computers to address one another by name and establish sessions for other protocols, such as SMB. This uses UDP port 137 for name services and TCP port 139 for session services.

The File Transfer Protocol (FTP) allows clients to upload and download files from a network server. It uploads files to websites and utilizes TCP ports 20 and 21.

HyperText Transfer Protocol (HTTP) is usually used to serve HTML web pages or applications hosted on a web server and typically operates over TCP port 80.

Related Content

-  6.3.6 Well-Known Ports
-  7.1.1 File/Print Servers
-  7.1.9 Remote Terminal Access Servers
-  14.2.1 Remote Desktop Tools

resources\questions\pt_core1_220_1201_2_2_009.question.xml

Question 10

✔ Correct

A company has found that specific departments are more productive when working from home as remote workers. The company would like to implement this ability and has reached out to a network security firm. What foundational infrastructure solution should the company implement to allow remote workers to access all the needed resources from their laptops at home?

☒ Virtual private network ✔ Correct

☐ Mail exchange records

☐ Virtual LAN

☐ DNS (AAAA) records

Explanation

A virtual private network (VPN) enables hosts to connect to the LAN without being physically installed at the site. This would be the first foundational step to implement in the given scenario.

A virtual LAN (VLAN) is a logical network segment using a feature of managed switches to segregate network traffic. Though the company may use this as a layer of security, it would not be foundational.

A mail exchange (MX) record identifies an email server for the domain so that other servers can send messages to it.

In the Domain Name System (DNS), an AAAA record resolves a hostname to an IPv6 address. This would not help with granting remote workers access to resources.

Related Content

6.4.7 Virtual Private Networks

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Question 11

✔ Correct

This protocol makes a secure connection to display the graphical interface of another computer and uses TCP port 3389.

- ☐ Server Message Block
- ☐ Simple Network Management Protocol
- ☒ Remote Desktop Protocol ✓ Correct
- ☐ Trivial File Transfer Protocol

Explanation

Remote Desktop Protocol (RDP) makes a secure connection to the graphical desktop of a remote computer and uses TCP port 3389.

Simple Network Management Protocol (SNMP) queries status information from network devices and uses UDP port 161. SNMP traps information sent to a management server utilizing UDP port 162.

Server Message Block (SMB) is the application protocol underpinning file and printer sharing on Windows networks. SMB usually runs directly over TCP port 445.

Trivial File Transfer Protocol (TFTP) is a simplified form of FTP supporting only file copying and uses UDP port 69.

Related Content

-  6.3.6 Well-Known Ports
-  7.1.1 File/Print Servers
-  7.1.9 Remote Terminal Access Servers
-  14.2.1 Remote Desktop Tools

resources\questions\pt_core1_220_1201_2_1_003.question.xml

Question 12

✔ Correct

This can be used individually or in conjunction with other configurations to effectively verify email services and block the transmission of spoofed and unwanted messages. (Select all that apply.)

☒ Sender Policy Framework ✔ Correct

☒ DomainKeys Identified Mail ✔ Correct

☒ Domain-based Message Authentication, Reporting, and Conformance ✔ Correct

☐ Virtual private network

Explanation

The Sender Policy Framework (SPF) record -there must be only one per domain- identifies the hosts authorized to send email from that domain.

DomainKeys Identified Mail (DKIM) uses cryptography to validate the source server for a given email message. This can replace or supplement SPF.

The Domain-based Message Authentication, Reporting, and Conformance (DMARC) framework ensures that the developer effectively utilizes SPF and DKIM. A DMARC policy publishes as a domain name system (DNS) TXT record.

A virtual private network (VPN) enables hosts to connect to the LAN without being physically installed at the site.

Related Content

6.4.5 DNS Spam Management Records

resources\questions\pt_core1_220_1201_2_4_002.question.xml

Question 13

✕ Incorrect

A company meets with an IT consultant about building a small data center within their facility. The company would house all its servers in this data center, and each server would have local disks set up for all the servers to share over the network. The consultant would question this method and suggest installing which storage solution into the data center?

☐ SAN ✓ Correct☒ LAN ✕ Incorrect☐ PAN☐ SCSI**Explanation**

A storage area network (SAN) provisions access to a configurable pool of storage devices that application servers can use, isolated from the main network. It is only accessed by servers, not by client PCs and laptops. This would be the best solution for a data center build.

A local area network (LAN) consists of computers connected by cabling and one or more network switches installed at a single geographical location.

Small Computer System Interface (SCSI) is a legacy standard for physically connecting and transferring data between computers and peripheral devices.

A personal area network (PAN) refers to using wireless connectivity to connect to devices at a range of a few meters.

Related Content

5.1.1 LANs and WANs

resources\questions\pt_core1_220_1201_2_7_003.question.xml

Question 14

✔ Correct

An IPv6 address is composed of which two main parts? (Select two.)

☐ Dynamic address

☒ Network ID ✔ Correct

☐ Static address

☒ Interface ID ✔ Correct

Explanation

An IPv6 address is in two main parts. The first 64 bits are a network ID.

An IPv6 address is in two main parts. The second 64 bits designate a specific interface.

Using static addressing requires an administrator to visit each computer to enter the configuration information for that host manually. This is traditionally only used for IPv4 addresses.

A host can receive its IP address, subnet mask, default gateway, and DNS server addresses from a Dynamic Host Configuration Protocol (DHCP) server as an alternative to static configuration. This is also used traditionally with only IPv4 addresses.

Related Content

6.2.15 IPv6 Addressing

resources\questions\pt_core1_220_1201_2_6_002.question.xml

Question 15

✕ Incorrect

A developer asks that any network requests made to the hostname dev23.company.net get directed to a specific IP address of 10.130.25.15. What type of record will the developer need to create?

- ☐ (TXT) record
- ☐ (MX) record
- ☒ (AAAA) record ✕ Incorrect
- ☐ (A) record ✓ Correct

Explanation

In the Domain Name System (DNS), an A record resolves a hostname to an IPv4 address. This will be the record type that the developer will need to create on the DNS server.

In the Domain Name System (DNS), an AAAA record resolves a hostname to an IPv6 address. The IP address in this scenario is IPv4.

A mail exchange (MX) record will identify an email server for the domain so that other servers can send messages to it.

A TXT record stores any free-form text that they may need to support other network services. They are most commonly used to verify email services and block the transmission of spoofed and unwanted messages.

Related Content[6.4.4 DNS Record Types](#)[resources\questions\pt_core1_220_1201_2_4_001.question.xml](#)

Question 16

✔ Correct

A company asked a network engineer to provide a solution that will allow a hosted application to be highly available if one or more servers are offline. They have plenty of resources to provide a cluster of servers that can run the application. Still, the company is looking for direction on making the application seamlessly available to customers. What solution would the engineer implement for this request?

- ☐ Proxy server
- ☐ Web server
- ☒ Load balancer ✓ Correct
- ☐ Spam gateway

Explanation

A load balancer can distribute client requests across server nodes in a farm or pool. The company can use a load balancer in any situation where multiple servers provide the same function.

A proxy server takes a full HTTP request from a client, checks it, and then forwards it to the destination server on the internet. This activity would not be the solution for the given request.

A web server provides client access using HTTP or its secure version (HTTPS). This is likely the type of servers in the mentioned cluster hosting the application.

Spam gateways verify the authenticity of mail servers and have filters that can identify spoofed, misleading, malicious, or otherwise unwanted messages.

Related Content 7.2.1 Proxy Servers 7.2.4 Load Balancers

resources\questions\pt_core1_220_1201_2_3_004.question.xml

Question 17

✔ Correct

A contractor working for a managed services company works fully remote from their residence. They have multiple desktops, printers, and even some Wi-Fi access points set up. What type of network is this considered?

☒ SOHO ✔ Correct

☐ WLAN

☐ WAN

☐ MAN

Explanation

A small office home office (SOHO) LAN is a business-oriented network, potentially using a centralized server in addition to client devices and printers, but often using a single networking appliance to provide LAN and Internet connectivity.

A wide area network (WAN) spans multiple geographic locations. One example of a WAN is the internet, a global network of networks.

A wireless local area network (WLAN) uses radios and antennas for data transmission and reception. Though there are some Wi-Fi access points, not all devices are wireless.

A metropolitan area network (MAN) can mean a specific network type covering an area equivalent to a city or other municipality.

Related Content

5.1.1 LANs and WANs

resources\questions\pt_core1_220_1201_2_7_002.question.xml

Question 18

✔ Correct

A network engineer is designing an enterprise network for their firm's new office. This focus on security will require the engineer to consider what fundamental concept when combining wired and wireless devices in an enterprise local area network?

- ☐ IEEE 802.11
- ☒ Segmentation ✓ Correct
- ☐ Wi-Fi
- ☐ Firewall

Explanation

Segmentation separates devices in various ways to better manage network traffic. Personal wireless devices are constantly coming and going, and it is best to have that section of a network segregated from the rest of the LAN.

IEEE 802.11 is better known by its brand name, Wi-Fi. This is just a set of standards used in implementing a wireless local area network (WLAN).

Wi-Fi is the brand name for the IEEE 802.11 standards used to implement wireless local area networks (WLANs).

Though some companies may use firewall solutions between internal network segments, this is not typical. Firewall technologies can give network engineers granular control over network traffic flow within an enterprise LAN.

Related Content

resources\questions\pt_core1_220_1201_2_7_004.question.xml

Question 19

✔ Correct

A network technician installs wireless access points (APs) in an office and would like to use power over Ethernet (PoE) to power the APs. The current network infrastructure is not capable of doing this. What would the network technician need to install to augment the power requirements for PoE APs?

- ☐ Patch panel
- ☒ Power injector ✓ Correct
- ☐ Firewall
- ☐ Managed switch

Explanation

If a switch does not support PoE, a network technician can use a power injector. One port on the injector connects to the switch port. The other port connects to the device.

Firewalls have rules that list source and destination network addresses, protocol types, and whether to allow or block traffic matching the rule. This would not provide power to connected devices.

Managed switches have internal software for more advanced functionality. Though managed switches can provide power, this scenario points out that it is not available.

In most types of office cabling, the computer connects to a wall port, and via cabling running through the walls, to a patch panel. Patch panels do not provide inline power to connected devices.

Related Content[5.2.10 Power over Ethernet](#)[resources\questions\pt_core1_220_1201_2_5_007.question.xml](#)

Question 20

✔ Correct

A user reports that their PC can not access the internet and internal network resources. The user also reports that the internet is intermittent throughout the day. A technician confirmed that the Ethernet cable is connected securely to the PC and the wall port. What tool should an engineer bring with them to investigate this further?

- ☐ Cable stripper
- ☐ Punchdown tool
- ☒ Cable tester ✔ Correct
- ☐ Snips

Explanation

A cable tester is a pair of devices that attach to each end of a cable. It tests a patch cord or connects via patch cords to a wall port and patch panel port to test the permanent link.

A punchdown tool fixes each conductor into an insulation displacement connector (IDC). There is no punching down wires into a block in this scenario. This will only be required if the cable tester shows a fault in the permanent link.

A cable stripper scores the outer jacket just enough to remove it.

Snips would cut the cable and trim the ripcord and the start filler inside the cable, should this process be necessary after testing the patch cable and permanent link.

Related Content

5.3.8 Copper Cabling Test Tools

resources\questions\pt_core1_220_1201_2_8_002.question.xml